Imitation Learning: BC and DAgger

1 Recovery under distribution shift

Use discounted *costs*, with $0 < \gamma < 1$ and $0 \le \epsilon \le 1$. Let C denote cost starting safe and D cost starting in the ditch. On a safe step, a correct action costs 0; a mistake costs 1 immediately. Every ditch action costs 1, including a successful recovery action.

1.1 Behavior cloning

BC never recovers after entering an unfamiliar ditch state. Thus

$$D_{\mathrm{BC}} = 1 + \gamma D_{\mathrm{BC}} = \frac{1}{1-\gamma},$$
$$C_{\mathrm{BC}} = (1-\epsilon)\gamma C_{\mathrm{BC}} + \epsilon(1 + \gamma D_{\mathrm{BC}}).$$

Since $1 + \gamma D_{\mathrm{BC}} = 1/(1-\gamma)$, rearranging gives the exact answer

$$\boxed{J(\pi_{\mathrm{BC}}) = C_{\mathrm{BC}} = \frac{\epsilon}{(1-\gamma)(1-\gamma+\gamma\epsilon)}.}$$

This is at most $1/(1-\gamma)$, as expected because every step costs at most 1.

1.2 Dataset aggregation

DAgger's recovery succeeds with probability $1-\epsilon$. Its two equations are

$$C_{\mathrm{D}} = \gamma(1-\epsilon)C_{\mathrm{D}} + \epsilon(1 + \gamma D_{\mathrm{D}}),$$
$$D_{\mathrm{D}} = 1 + \gamma\big[(1-\epsilon)C_{\mathrm{D}} + \epsilon D_{\mathrm{D}}\big].$$

Subtracting the first equation from the second yields $D_{\mathrm{D}} - C_{\mathrm{D}} = 1 - \epsilon$. Substitution into the first equation gives

$$(1-\gamma)C_{\mathrm{D}} = \epsilon\big[1 + \gamma(1-\epsilon)\big], \qquad \boxed{J(\pi_{\mathrm{DAgger}}) = \frac{\epsilon(1+\gamma-\gamma\epsilon)}{1-\gamma}.}$$

1.3 When the familiar scaling is valid

Writing $\delta = 1-\gamma$, BC's denominator is $\delta(\delta + \gamma\epsilon)$. The approximation $C_{\mathrm{BC}} \approx \epsilon/\delta^2$ needs $\gamma\epsilon \ll \delta$, not merely $\epsilon \ll 1$ and $\delta \ll 1$. For example, if $\epsilon = \delta$ and $\gamma \to 1$, neglecting the second term in $\delta + \gamma\epsilon$ loses a factor approaching 2.

For small ϵ, DAgger has $C_{\mathrm{D}} \approx \epsilon(1+\gamma)/(1-\gamma)$. In the regime $\epsilon \ll 1-\gamma$, BC therefore exhibits a squared effective-horizon factor, while recovery gives DAgger a linear factor. These are properties of this toy MDP, not universal exact formulas for all imitation-learning tasks.

2 Solving the cost-minimization MDP

For a deterministic transition $s' = f(s, a)$,

$$Q^*(s, a) = c(s, a) + \gamma V^*(f(s, a)), \qquad V^*(s) = \min_a Q^*(s, a).$$

The self-loops are mandatory and do not terminate the return. At $\gamma = 0.9$,

$$V^*(3) = \frac{1}{1 - 0.9} = 10, \quad V^*(4) = 0, \quad V^*(5) = \frac{1.5}{1 - 0.9} = 15.$$

Work backward through the two decision states:

$$V^*(2) = \min\{18 + 0.9V^*(4),\ 0 + 0.9V^*(5)\} = \min\{18, 13.5\} = 13.5,$$
$$V^*(1) = \min\{0 + 0.9V^*(2),\ 2 + 0.9V^*(3)\} = \min\{12.15, 11\} = 11.$$

State	$V^*(s)$	Optimal action
1	11	a_2, move to 3
2	13.5	a_2, move to 5
3	10	a_1, self-loop
4	0	a_1, self-loop
5	15	a_1, self-loop

State 1 illustrates why immediate cost is insufficient: paying 2 now is better than paying 0 and then following the more expensive continuation.

3 Programming solution

For continuous actions, behavior cloning minimizes the supervised loss

$$L_{\mathrm{BC}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left\| \pi_\theta(o_i, g_i) - a_i^E \right\|_2^2.$$

Concatenate proprioception and goal along the last axis, use an MLP, and bound the action with a hyperbolic tangent. Each training batch clears gradients, computes predictions and MSE, backpropagates, and steps Adam. Evaluate a fixed held-out set without gradients and select the checkpoint by validation loss.

DAgger starts from the BC dataset and repeats:

1. Execute the current learner to visit its own state distribution.
2. At each visited state, ask the expert for the target action.
3. Add the state and *expert* label to the accumulated training set.
4. Retrain on all accumulated data and evaluate the resulting policy.

Executing the learner while labeling with the expert is the key distinction. Storing learner actions as targets would reinforce its mistakes. Recovery depends on the expert being able to label the visited states and on the learner being able to represent the needed behavior.

4 Reading the retained experiment

Experiment status. The following numbers and curves are retained from earlier runs, not from a rerun of the revised reference code. They are examples for interpretation, not performance guarantees.

Policy	Mean episode return
Random actions	−46.2
Expert	−1.8
Behavior cloning	−11.5
DAgger	−2.3

These runs use the sparse reaching reward: unsuccessful steps incur −1. Less-negative returns indicate faster successful reaching under the same evaluation convention. The random and expert rollouts used 10 episodes; learner evaluation used 20. These small, unequal samples do not establish a statistically reliable ranking.

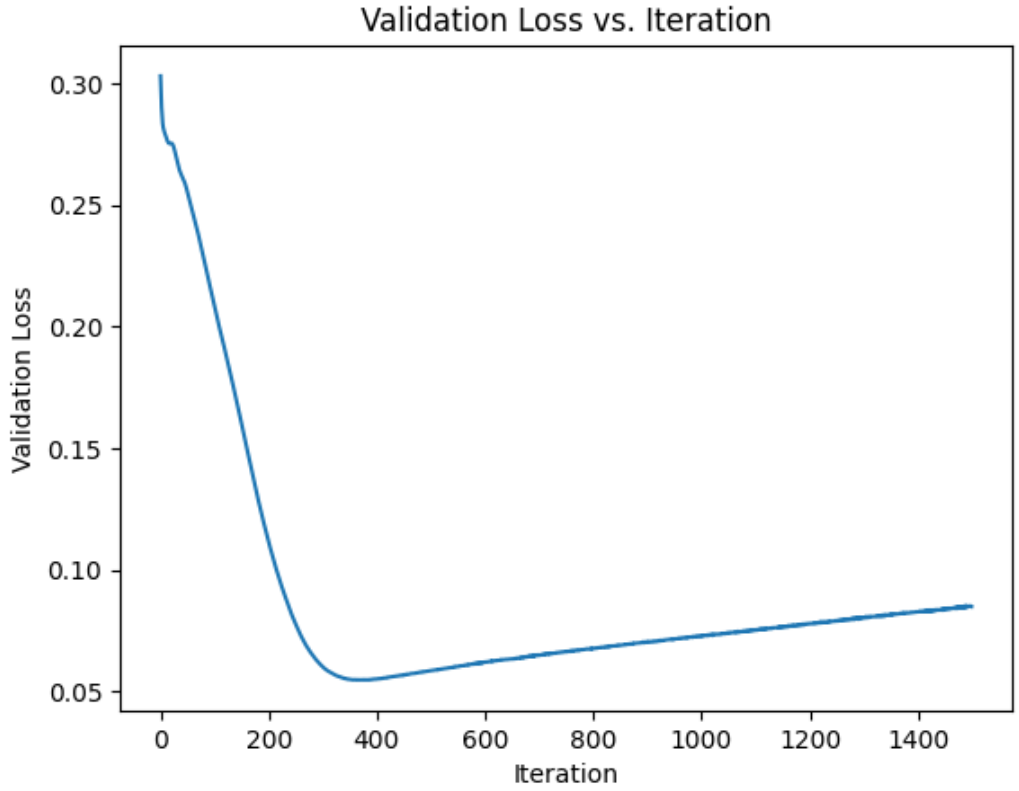


Figure 1: Retained BC validation curve. The late increase motivates selecting a validation checkpoint rather than simply using the final epoch.

The curve falls initially and later rises. That behavior is consistent with overfitting; a low supervised loss alone also does not guarantee robust closed-loop control outside the expert distribution. DAgger targets that distribution mismatch by collecting labels on learner-visited states.

The revised reference uses the local shared expert checkpoint and no longer downloads it from an external exercise repository. Its checkpoint selection uses held-out loss. Results must be measured again after these code changes.

Solution scope. The exact discounted costs and MDP values follow from the specified transitions. The empirical explanation is an interpretation of historical observations, not a proof of an algorithm-wide performance gain.